

N100 Financial Intelligence Platform

Analyst Guide: Dashboard, Reports & API Reference

This guide covers day-to-day usage of the platform: navigating the Streamlit dashboard, running the screener, generating PDF tearsheets, calling the REST API directly, and resolving the handful of issues that come up most often.

1. Getting Started

The platform has three parts you'll interact with day to day: the Streamlit dashboard (for browsing and screening), the tearsheet/report PDFs (for sharing a company's numbers outside the app), and the REST API (for pulling data into your own scripts or tools).

>> Initiate Streamlit UI:

```
streamlit run src/dashboard/app.py
```

Run this from the project root, not from inside src/dashboard/ — the data loader resolves file paths relative to the project root and will fail to find the database otherwise.

>> Starting the FastAPI Server:

```
uvicorn src.api.main:app --port 8000 --reload
```

Once running, open <http://127.0.0.1:8000/docs> for interactive OpenAPI documentation — every endpoint below can be tried directly from that page.

2. Dashboard Screens

2.1 Home

Six portfolio-wide KPI tiles (winsorized average ROE, median P/E, median D/E, total companies, median revenue CAGR, debt-free count), a sector breakdown donut chart, and a Top-5 companies table by composite quality score. A year selector (2019–2024) drives the KPI tiles.

2.2 Company Profile

Search any of the 92 companies by name or ticker. Shows a company card (sector, sub-sector, description), 6 KPI tiles, a 10-year Revenue/Net Profit bar chart, an ROE trend line, and pros/cons badges where available. Note: ROCE is shown as a flat reference line, not a trend — see Troubleshooting §5.3.

2.3 Screener

10 sliders (ROE, D/E, FCF, Revenue CAGR, PAT CAGR, OPM, P/E, P/B, Dividend Yield, ICR) filter the universe live. 6 preset buttons pre-fill the sliders to approximate common screens. Results export to CSV. This screen calls the exact same underlying function (`run_screener()`) as the API's `/screener` endpoint — the two are guaranteed to agree (see `tests/api/test_integration.py`).

2.4 Peer Comparison

Pick one of 11 peer groups and a company within it to see an 8-axis radar chart against the group average, plus a full KPI table with the benchmark company highlighted.

2.5 Trend Analysis

Overlay up to 3 metrics for one company over its available history, with YoY % change annotated at each point.

2.6 Sector Analysis

A Revenue-vs-ROE bubble chart (bubble size = market cap) for a selected sector, plus sector median KPIs. Note: there are 10 sectors in this dataset, not the 11 sometimes referenced in older planning notes.

2.7 Capital Allocation Map

A treemap of all 92 companies grouped into 8 capital-allocation patterns (Reinvestor, Shareholder Returns, Growth Funded by Debt, etc.), derived from each company's latest operating/investing/financing cash flow signs. Click a pattern or company to drill in.

2.8 Annual Reports

Search a company to see its available report years with links to BSE; links that don't resolve to a real `http(s)` address are flagged in red.

3. Generating PDF Reports

3.1 Company tearsheets

Tearsheets are pre-generated, not built on the fly — this keeps the dashboard and API fast. To regenerate all of them (e.g. after a data refresh):

```
python -m src.reports.tearsheet batch
```

This produces one 2-page PDF per company in reports/tearsheets/, skips any company with fewer than 3 years of history (logged to output/skipped_tearsheets.csv — currently just JIOFIN), and logs any genuine failure separately to output/tearsheet_failures.csv.

To regenerate just the 5-company spot-check used during development:

```
python -m src.reports.tearsheet
```

3.2 Sector reports

```
python -m src.reports.sector_report
```

Produces one PDF per sector (10 total) in reports/sector/.

3.3 Portfolio summary

```
python -m src.reports.portfolio_summary
```

Produces a single 92-page PDF (one page per company, alphabetical by ticker) in reports/portfolio/, with a trend arrow per KPI versus the prior year.

3.4 Fetching a tearsheet via the API

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/tearsheet -o TCS_tearsheet.pdf
```

4. Calling the API

All 16 endpoints live under `/api/v1`. Full request/response schemas are in `docs/openapi.json`, or importable into Postman via `docs/postman_collection.json`. The most commonly used ones, with example calls:

>> Health check

```
curl http://127.0.0.1:8000/api/v1/health
```

>> List all companies

```
curl http://127.0.0.1:8000/api/v1/companies
```

>> Filter companies by sector

```
curl "http://127.0.0.1:8000/api/v1/companies?sector=Information Technology"
```

>> Company profile

```
curl http://127.0.0.1:8000/api/v1/companies/TCS
```

>> P&L; history (year-bounded)

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS/pl?from_year=2022-03&to_year=2024-03"
```

>> Ratios for one year

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS/ratios?year=2024-03"
```

>> Screener

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1.0"
```

>> Sector list with medians

```
curl http://127.0.0.1:8000/api/v1/sectors
```

>> Companies in one sector

```
curl "http://127.0.0.1:8000/api/v1/sectors/Information Technology/companies"
```

>> Peer group percentiles

```
curl "http://127.0.0.1:8000/api/v1/peers/IT Services"
```

>> Radar comparison vs peers

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/peers/compare
```

>> Valuation history

```
curl http://127.0.0.1:8000/api/v1/market-cap/TCS
```

>> Portfolio-wide percentiles

```
curl http://127.0.0.1:8000/api/v1/portfolio/stats
```

>> Annual report links

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/documents
```

5. Troubleshooting

5.1 “Company not found” (404)

Check the ticker spelling and case — tickers are exact-match (e.g. TCS, not tcs or Tcs). Use GET `/api/v1/companies?search=...` to find the right ticker if unsure.

5.2 A company's tearsheet 404s but the company itself exists

That company was skipped during batch tearsheet generation for having fewer than 3 years of financial history — check `output/skipped_tearsheets.csv`. As of this writing only JIOFIN is affected. This is expected behavior, not a bug.

5.3 ROCE looks flat / doesn't match the tearsheet's ROCE line

There is no per-year ROCE anywhere in the source data — only a single latest-year value (`companies.roce_percentage`). Anywhere a ROCE trend is shown, it's a derived proxy (EBIT / Capital Employed) and is always labelled “(derived)”. Treat it as an approximation, not an official figure.

5.4 The screener returns fewer than 92 companies with no filters set

This is expected: SBIN and ATGL have zero rows in `financial_ratios` (SBIN because D/E-style ratios don't meaningfully apply to a bank; ATGL for an unresolved upstream data gap), so they don't appear in the screener's universe at all. They still appear normally everywhere else (tearsheets, portfolio summary, pros/cons) — this is specific to the screener.

5.5 A screener filter (e.g. `max_de`) doesn't seem to apply to some companies

Some filters are intentionally exempt for the Financials sector — `max_de`, for example, skips banks/NBFCs, since a high D/E is structurally normal for them, not a quality flag. This is configured in `config/screener_config.yaml`'s `skip_sector` setting, not a bug.

5.6 Sector name doesn't match what I expected (e.g. “IT”)

Sector names are the exact `broad_sector` values in the database — “Information Technology”, not “IT”.

Use GET `/api/v1/sectors` to see the exact 10 valid names.

5.7 Port already in use when starting the dashboard or API

The dashboard defaults to port 8501, the API to 8000 — they don't conflict with each other. If you see an “address already in use” error, another instance is likely already running; stop it first or pass a different `--port` / `--server.port`.

5.8 Where to find known data gaps

The project README's “Known data limitations” section and each sprint's retrospective document (`docs/sprint*_Retrospective.md`) list every gap found during development, why it exists, and how it's handled — check there before assuming something is broken.

6. KPI Glossary

Every metric shown across the dashboard, tearsheets, and API, in one place.

Metric	Definition	Where it's computed
ROE %	Net Profit / Shareholder Equity. Returns None when equity is negative rather than a misleading negative-of-negative ratio.	financial_ratios
ROCE %	EBIT / Capital Employed. A single latest-year snapshot in companies table; anywhere shown as a trend, it's a derived proxy — see §5.3.	companies / derived
Debt/Equity	Total Borrowings / Shareholder Equity. 0 for debt-free companies.	financial_ratios
OPM %	Operating Profit / Sales.	financial_ratios
Net Profit Margin %	Net Profit / Sales.	financial_ratios
Interest Coverage (ICR)	EBIT / Interest Expense. None when interest is 0 (no meaningful ratio to compute).	financial_ratios
Revenue / PAT / EPS CAGR 5yr	5-year compound annual growth rate. Can be negative; that's a real declining business, not an error.	financial_ratios
Free Cash Flow (FCF)	Cash from Operations – CapEx. Allowed to be negative by design.	cashflow_kpis.py
FCF CAGR 5yr	5-year CAGR of FCF. Returns None when the start or end year's FCF is ≤ 0 (CAGR is mathematically undefined there).	cashflow_kpis.py
CFO Quality Score	CFO / PAT, averaged over up to 5 years. >1.0 = High Quality, $0.5\text{--}1.0$ = Moderate, <0.5 = Accrual Risk.	cashflow_kpis.py
CapEx Intensity	$ \text{Investing Activity} / \text{Sales}$. $<3\%$ = Asset Light, $3\text{--}8\%$ = Moderate, $>8\%$ = Capital Intensive.	cashflow_kpis.py
Distress Signal	$\text{CFO} < 0$ AND $\text{CFF} > 0$ in the latest year. Not meaningful for banks/NBFCs — see the flag's sector caveat in output/distress_alerts.csv.	cashflow_kpis.py
Capital Allocation Pattern	One of 8 labels (Reinvestor, Shareholder Returns, Growth Funded by Debt, Distress Signal, Liquidating Assets, Mixed, Pre-Revenue, Cash Accumulator) from the sign of CFO/CFI/CFF plus the CFO/PAT ratio.	cashflow_kpis.py
Composite Quality Score	Weighted blend of the above, winsorized at P10/P90 to limit the influence of extreme outliers.	composite_score.py
Dividend Payout Ratio %	Dividends Paid / Net Profit. $>100\%$ means dividends are funded from reserves — flagged as unsustainable.	financial_ratios

7. Database Tables

13 tables in db/nifty100.db (not 10 — the schema grew past the original plan; see README “Known data limitations”).

Table	Rows	What it holds
companies	92	Core identity: name, sector links, description, latest ROCE
profitandloss	1,070	Sales, expenses, operating profit, net profit, EPS per company-year
balancesheet	1,140	Equity, borrowings, assets, liabilities per company-year
cashflow	1,056	Operating/investing/financing activity per company-year
financial_ratios	1,041	All computed ratios (ROE, D/E, CAGRs, etc.) per company-year
sectors	92	broad_sector (10 values), sub_sector, market cap category
market_cap	552	P/E, P/B, EV/EBITDA, dividend yield, 2019–2024
stock_prices	5,520	Historical daily/periodic price data
peer_groups	56	Which companies belong to which of the 11 peer groups
peer_percentiles	533	Precomputed percentile rank per company per peer group per metric
documents	1,457	Annual report links per company-year (columns are capitalized “Year”/“Annual_Report” — the one schema inconsistency)
analysis	20	Free-text CAGR figures — only 5 of 92 companies have any data here
prosandcons	14	Raw manually-sourced pros/cons — not used by the platform's own pros/cons generator, which computes independently from financial_ratios

7.1 A worked example: finding quality-but-cheap companies

A common analyst workflow: find financially strong companies the market hasn't fully priced in yet. Using the Screener page (or the API directly):

- Set $ROE \geq 15\%$ and $D/E \leq 1.0$ (quality filters)
- Set Revenue CAGR 5yr $\geq 10\%$ (genuine growth, not a one-off good year)
- Sort by Composite Quality Score, then eyeball P/E from the results for anything still reasonably valued relative to its sector median (Sector Analysis page, or GET /api/v1/sectors for the median P/E per sector to compare against)

Equivalent API call: `curl`

```
"http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1.0&min_rev_cagr_5yr=10"
```


8. Full API Endpoint Reference

All 16 endpoints. Every one is a GET request; parameters marked (path) are part of the URL itself, (query) are appended as ?key=value.

Endpoint	Parameters	Returns
/api/v1/health	—	Status, row counts for all 13 tables, uptime, version
/api/v1/companies	sector, market_cap_category, search (all query, optional)	List of companies with sector and ROE/ROCE
/api/v1/companies/{ticker}	ticker (path)	Full profile + latest-year KPIs
/api/v1/companies/{ticker}/pl	ticker (path); from_year, to_year (query, optional)	P&L; history
/api/v1/companies/{ticker}/bs	ticker (path); from_year, to_year (query, optional)	Balance sheet history
/api/v1/companies/{ticker}/cashflow	ticker (path); from_year, to_year (query, optional)	Cash flow history
/api/v1/companies/{ticker}/ratios	ticker (path); year (query, optional)	Computed ratios, full history or one year
/api/v1/companies/{ticker}/tearsheet	ticker (path)	PDF binary download
/api/v1/companies/{ticker}/documents	ticker (path)	Annual report links + validity flag
/api/v1/companies/{ticker}/peers/compare	ticker (path)	Radar data vs peer group + benchmark
/api/v1/screener	min_roe, max_de, min_fcf, sector, min_rev_cagr_5yr, min_pat_cagr_5yr, max_pe (all query, optional)	Ranked companies by composite score
/api/v1/sectors	—	All 10 sectors with median ROE/PE/D-E
/api/v1/sectors/{sector_name}/companies	sector_name (path, exact match)	Companies in that sector
/api/v1/peers/{group_name}	group_name (path, exact match)	Companies in that peer group with percentile ranks
/api/v1/market-cap/{ticker}	ticker (path)	Valuation history, 2019–2024
/api/v1/portfolio/stats	—	P10–P90/mean/std for 10 KPIs across all 92 companies

9. Understanding This Platform's Data-Gap Labels

Throughout development, every genuine data gap or limitation found was documented rather than silently patched over or hidden. As an analyst using the platform day to day, you'll run into a few recurring labels — here's what each one actually means, so you can trust them rather than mistake them for bugs.

“(derived)”

Attached to any ROCE trend figure. Means it's computed from EBIT / Capital Employed as a proxy, not the platform's own official latest-year ROCE value — because no per-year ROCE exists anywhere in the source data. Use it for shape/direction, not as an exact number to quote externally.

“(N/A)”

A metric genuinely couldn't be computed for that company-year — most often because a denominator was zero or negative (e.g. ROE with negative equity, ICR with zero interest, CAGR from a negative base). This is intentionally different from showing a 0 or a fabricated number, which would be more misleading than admitting the gap.

“(FALLBACK)” (pros/cons only)

Means none of the 24 confidence-scored pro/con rules fired for that company (usually thin history), so the platform surfaced its single best/worst available metric instead, at a flat 61% confidence, so every company still has at least one pro and one con to show.

Sector-exempt filters

A few screener filters (like max D/E) are deliberately skipped for the Financials sector, because the underlying ratio doesn't mean the same thing for a bank as it does for a manufacturer. If a bank appears in your results despite a D/E filter, this is why.

For the full, itemized list of every gap found across all six sprints — with the reasoning behind each decision — see the project README's “Known data limitations” section and the individual sprint retrospective documents in docs/.